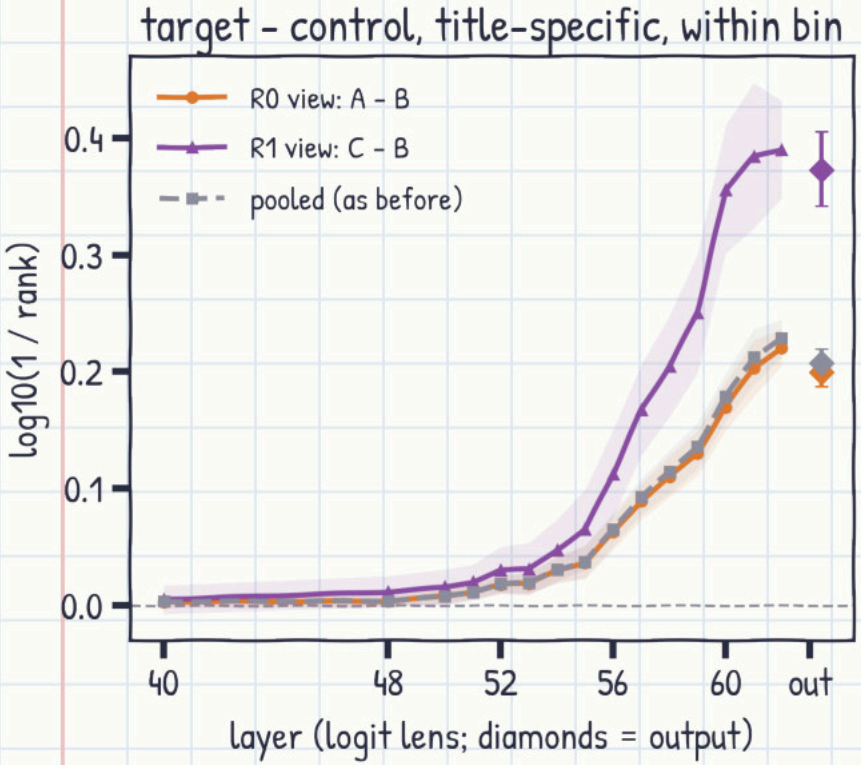
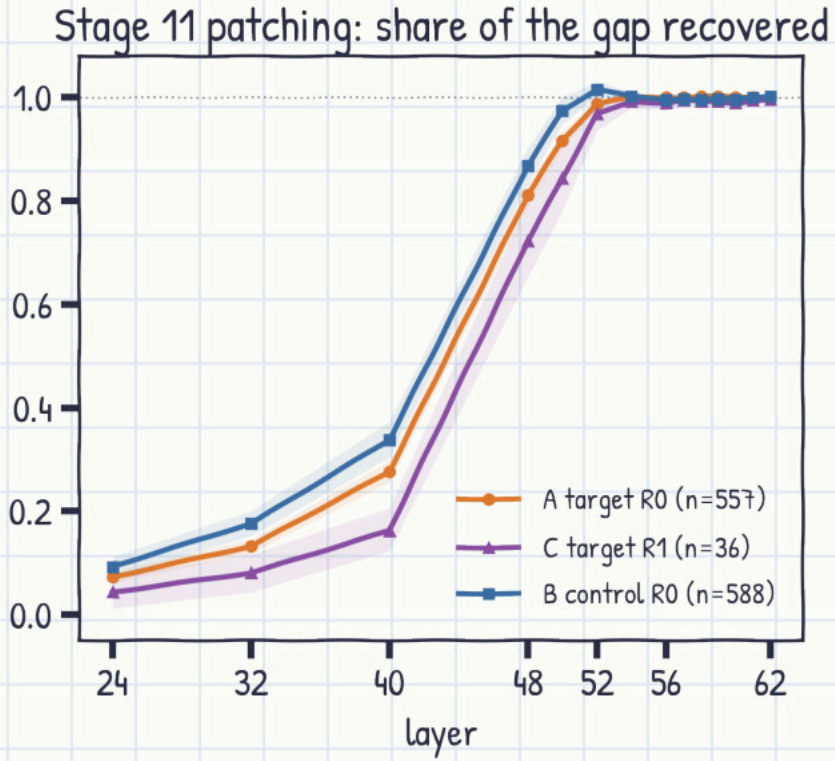
Stage 10–11

# Is it just the R1 papers? No.

R0 = no true author written · R1 = a few written · A/B/C/D = target/control x R0/R1




**R0 only (A – B):**

output +0.199 vs +0.207, same shape (Stage 10, A 3,423 / B 2,151, §6.13)

**R1 (C – B):** +0.37, also late (C 391 / B 885)

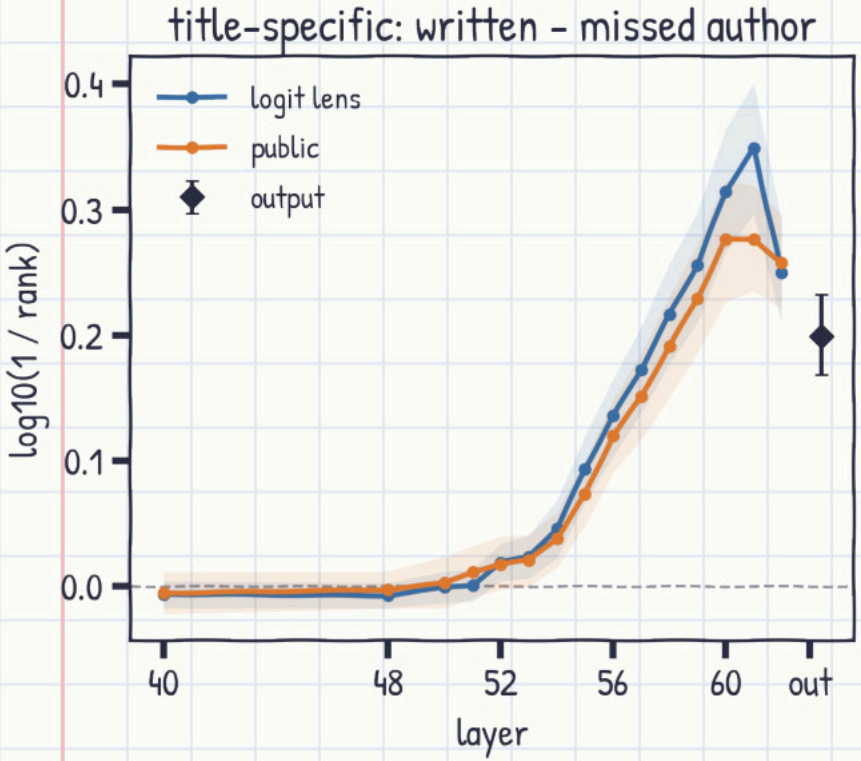
C – A within target: +0.11, from L56 (C 485 / A 3,204)

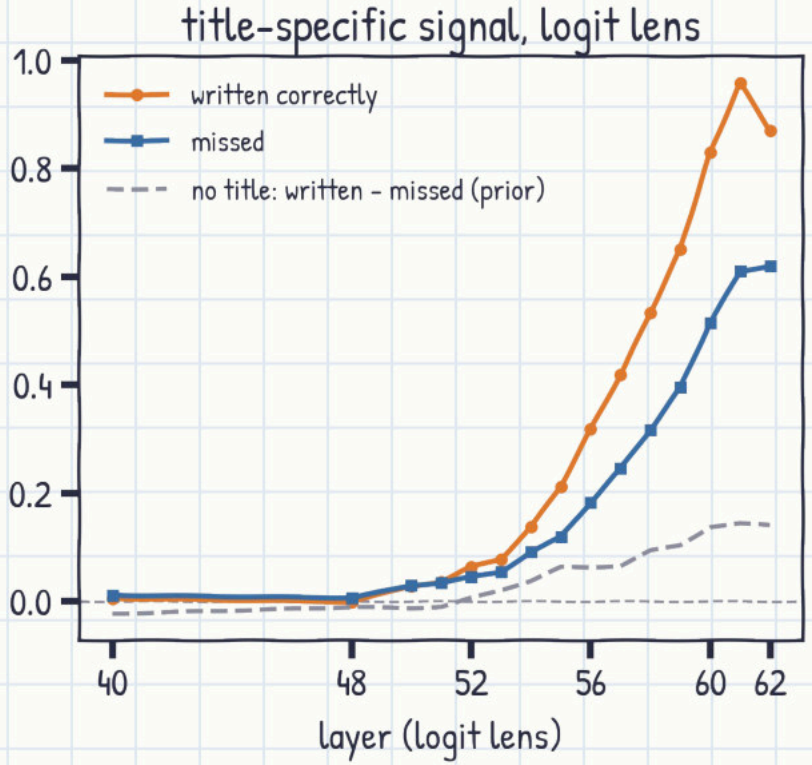
**patching:** R1 a bit later, done by L52 (Stage 11 E1, 36 R1 targets)

Stage 10–11

# Authors it wrote vs authors it missed

inside 495 R1 papers · paired within paper · 95% CI (Stage 10, r0_r1_analysis.md §4)




written − missed, output:
**+0.20** (Stage 10, final L63, 495 papers)

~0 up to L52, grows from **L54**

no title (prior): +0.10 (same, no title)

same without first authors

Stages 1–9 · 20–200 papers

# What we saw

- True title helps: pilot L56 +9.3 pp (7.3, 50 papers); 200 new papers +5.2 pp (8D RQ1, 200 papers)
- A–B uncertain in the pilot; +5.0 pp on 200 papers but gone after dropping 10 pairs (8D RQ2, 100 pairs)
- Low – high citation: inconclusive

Stages 10–11 · 9,108 papers

# What scale showed

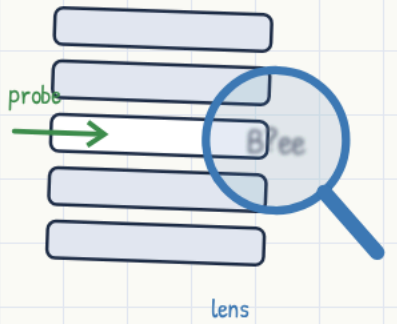
- Title-specific signal clear in every group, ≈0.44 at L62 (L62, §6.9)
- target – control ≈0 at L24–L40, slightly negative at L48 (−0.04 to −0.05) (§6.4), +0.21 [0.20, 0.22] at the output (L63, §6.4)
- Name prior (delete-title condition) is only ~0.35 of the gap; ~0 on surnames (§6.12)
- Signal rises steadily with citations ($\rho$ = 0.451) (§6.8)
- Same groups as the pilot (R0 only, A – B): +0.20 [0.19, 0.21] (R0, §6.13), same late shape
- R1 papers (C) know more (+0.11 over A) (§6.13) but are just as late

**Line A: from "the title helps" to "this paper, late, graded by fame"**

The pilot was right that the title matters. It could not see that the target advantage is real but confined to the last layers, so the 7-layer average and the L56 endpoint both diluted it. Restricting the full cohort to the pilot's A/B groups (R0 only) changes nothing: the R1 papers do not drive the result.

Line B

# What can each readout see?




small samples asked

Is J-Lens better than Vanilla? Is the fitted lens stable?

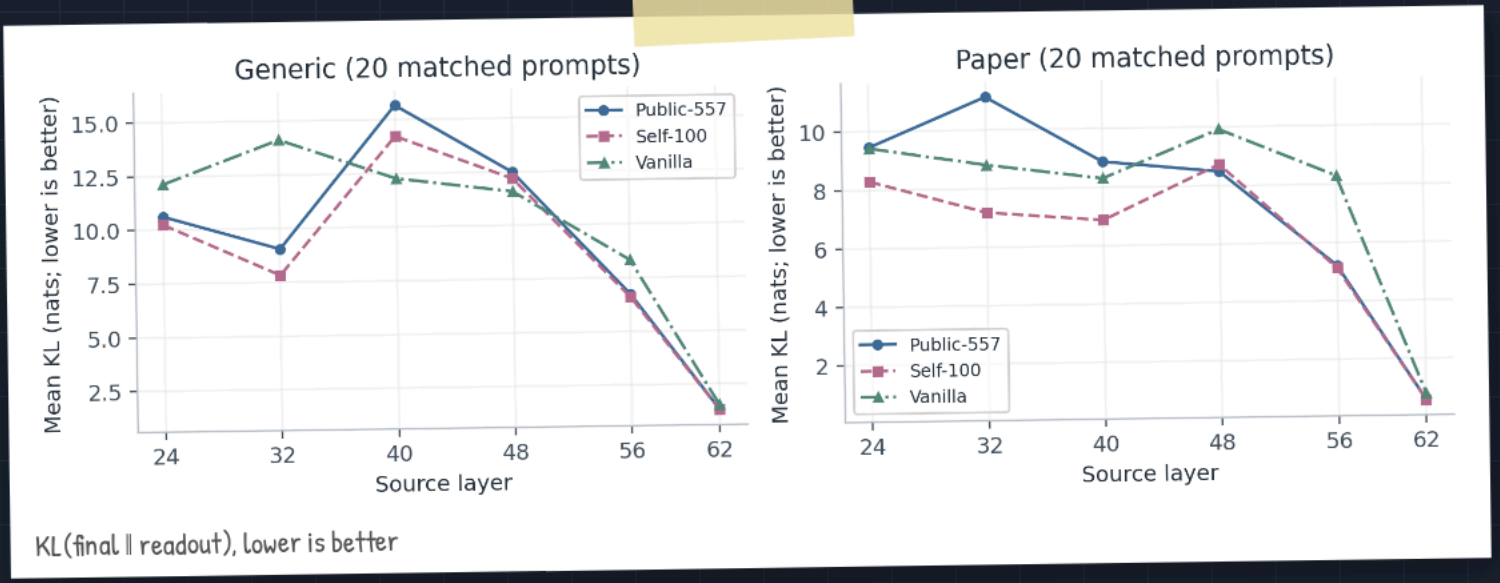
→

the full cohort answered

Lenses read name shape earlier, not paper identity; a probe at the title token reads paper-specific information

# Two lenses: Public-557 vs Self-100

Can a middle layer approximate the final next-token distribution? (20 generic + 20 paper prompts)

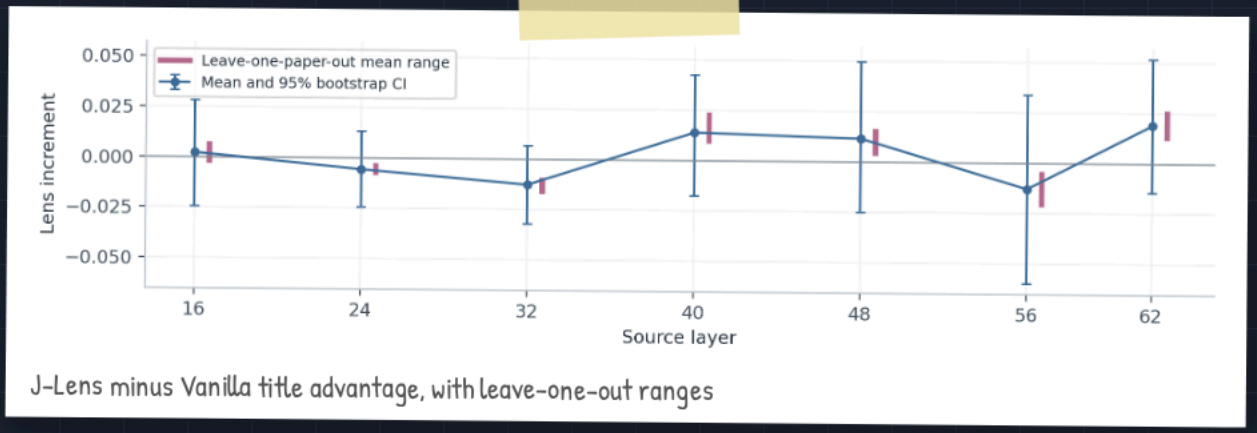



- Both checkpoints numerically valid (63/63 and 6/6 finite)(matrices.csv; frobenius_norms.csv)
- Self-100 has lower KL on 6/6 generic and 5/6 paper layers (paper L32: 11.1 → 7.2) (20 + 20 prompts; metrics.csv)
- But at paper L62, Self-100 KL beats Vanilla (0.58 vs 0.81) while Top-10 overlap is worse (0.36 vs 0.72)(metrics.csv)
- Author pilot: both lenses 14/20 full-name wins at L48(20 papers, layer_summary.csv)

Lesson: a better distribution match does not mean better author ranking. Fidelity cannot stand in for the task.

# Cross-stage audit: does J-Lens add value?

Read-only joins of Stages 5–7.3; 20,000 bootstrap draws




J-Lens minus Vanilla title advantage, with leave-one-out ranges

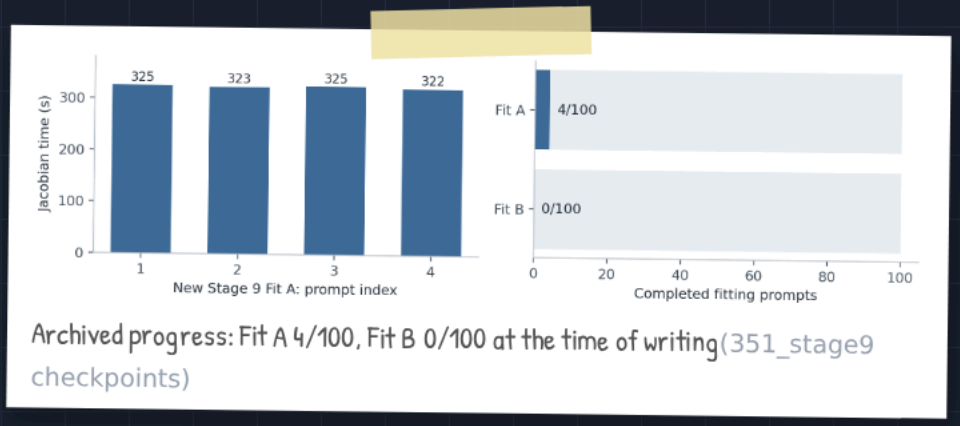
| | |
|---|---|
| Stage 5 vs 7.3 at L56 | r = 0.45, ρ = 0.47; signs agree 32/50 (50 papers, cross_stage_correlations.csv) |
| J-Lens − Vanilla | +0.002 [−0.008, +0.012]; median −0.002; 24/50 positive (jlens_increment_bootstrap.csv) |
| A−B of that increment | −0.014 [−0.033, +0.004] (20 pairs) |

Dropping one paper can flip the sign of the J-Lens increment. The pilot shows **no added title-specific value** from J-Lens over Vanilla. Everything so far is exploratory: L56 was inspected repeatedly.

# Stage 9: is the lens stable across fitting data?

New FullFit protocol, different from the historical 50+50 split

| | Historical split-half | Stage 9 FullFit |
|---|---|---|
| Fit data | Self-100 split 50 + 50 | two disjoint 100-prompt sets |
| Holdout | 20 generic + 20 paper | independent 100 WikiText |
| Layers | 6 | 7 (adds L16) |
| Status | complete | only Fit A completed; Fit B and holdout not run |




Archived progress: Fit A 4/100, Fit B 0/100 at the time of writing (351_stage9 checkpoints)

- WikiText-103 train, 100–1,200 chars, deduplicated by hash, seed 3519001
- ~324 s Jacobian time per prompt, ~62 GiB CPU RAM (full_A.log; benchmark log); per-prompt checkpoints

Only Fit A was finished, and Stage 10 used it as "fitA". Without Fit B and the holdout, fitting stability is **untested**.

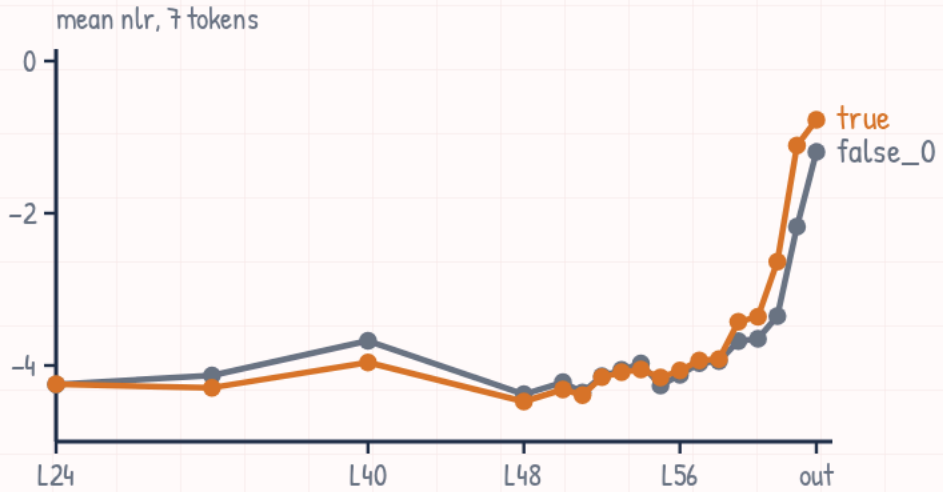
# Layer-by-layer readout

Coifman, Bonanno, Rafaeli (2007) · logit lens

1 prompt → 2 state at name → 3 lens → vocabulary → 4 rank of true token → 5 mean

**nlr = log10(1 / rank)**

rank = 1 + #{tokens scored above the true one}





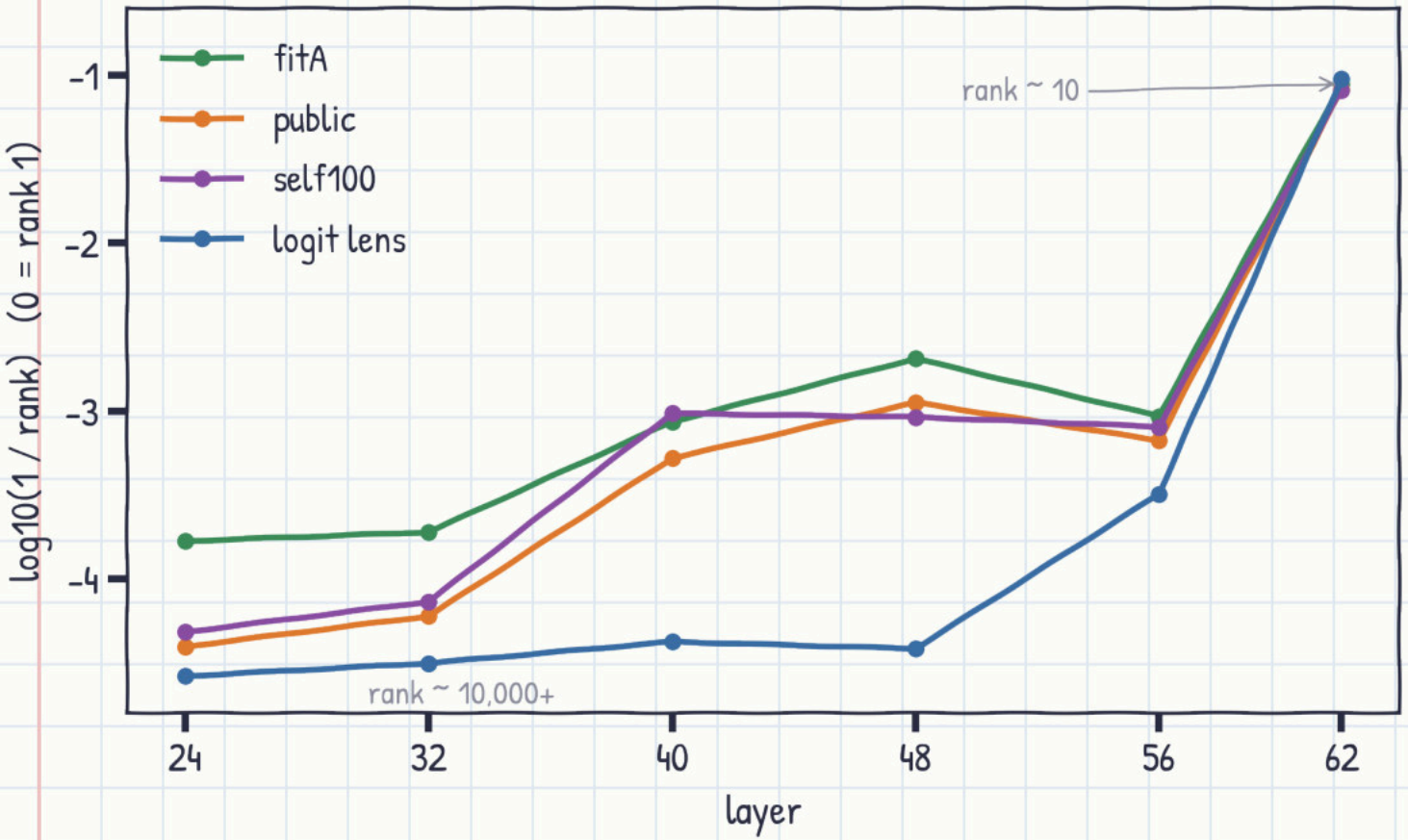


…titled | Affect | dynamics, | … | (2007)? | Authors | : | Kar | in | G | . | Co | if | man

Ranks of "Co" (author 1, token 5), true title: 351_stage10/raw/logit/L{24…62}/shard_011.npz row 25029; out = raw/final/L63

# Names become readable late

all 9,108 papers · true title (Stage 10 cohort, §6.1)




Logit lens: almost nothing until **L48**

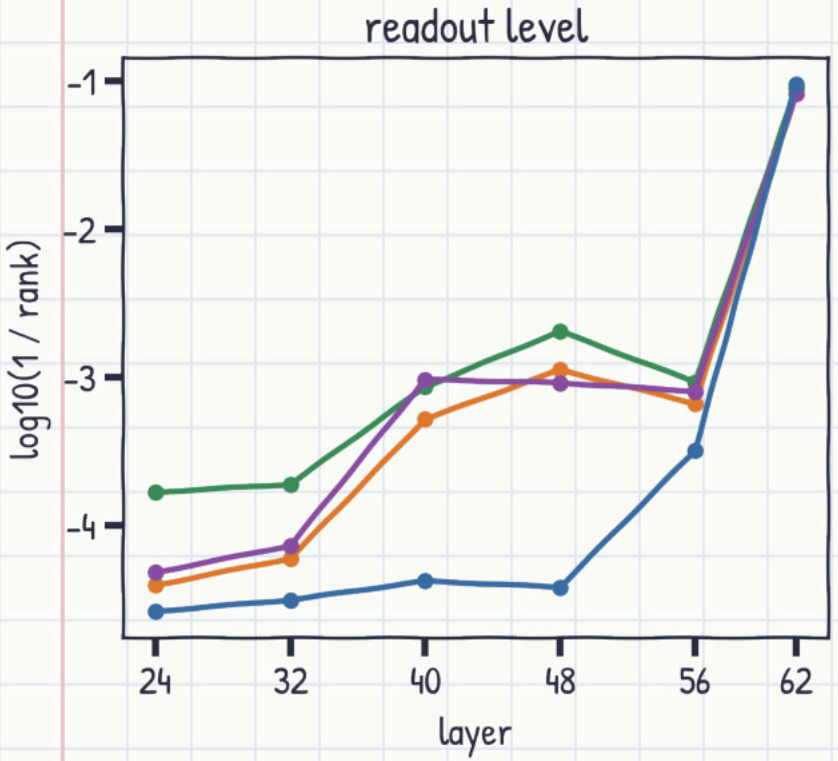
Jacobian lenses read more from **L40**

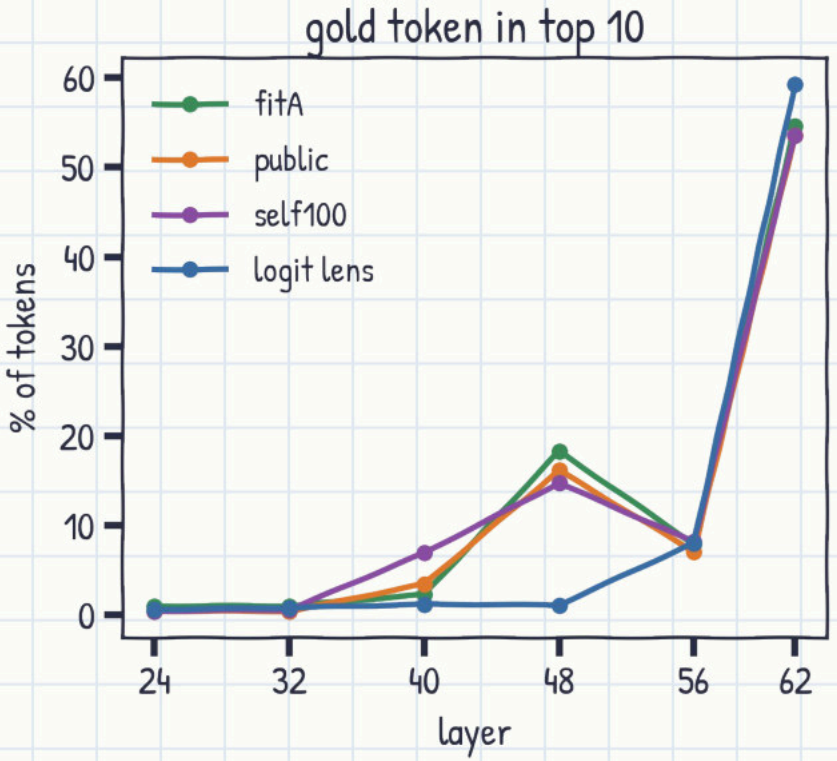
Everything jumps at **L62** (rank ~10) (Stage 10, logit L62 -0.93, §6.1)

Stage 10

# Which lens reads best?

all papers · true title




**Middle layers:**

fitA best; at L48 15–18% top-10 vs 1% for logit

(Stage 10 X3, 9,108 papers, §6.9)

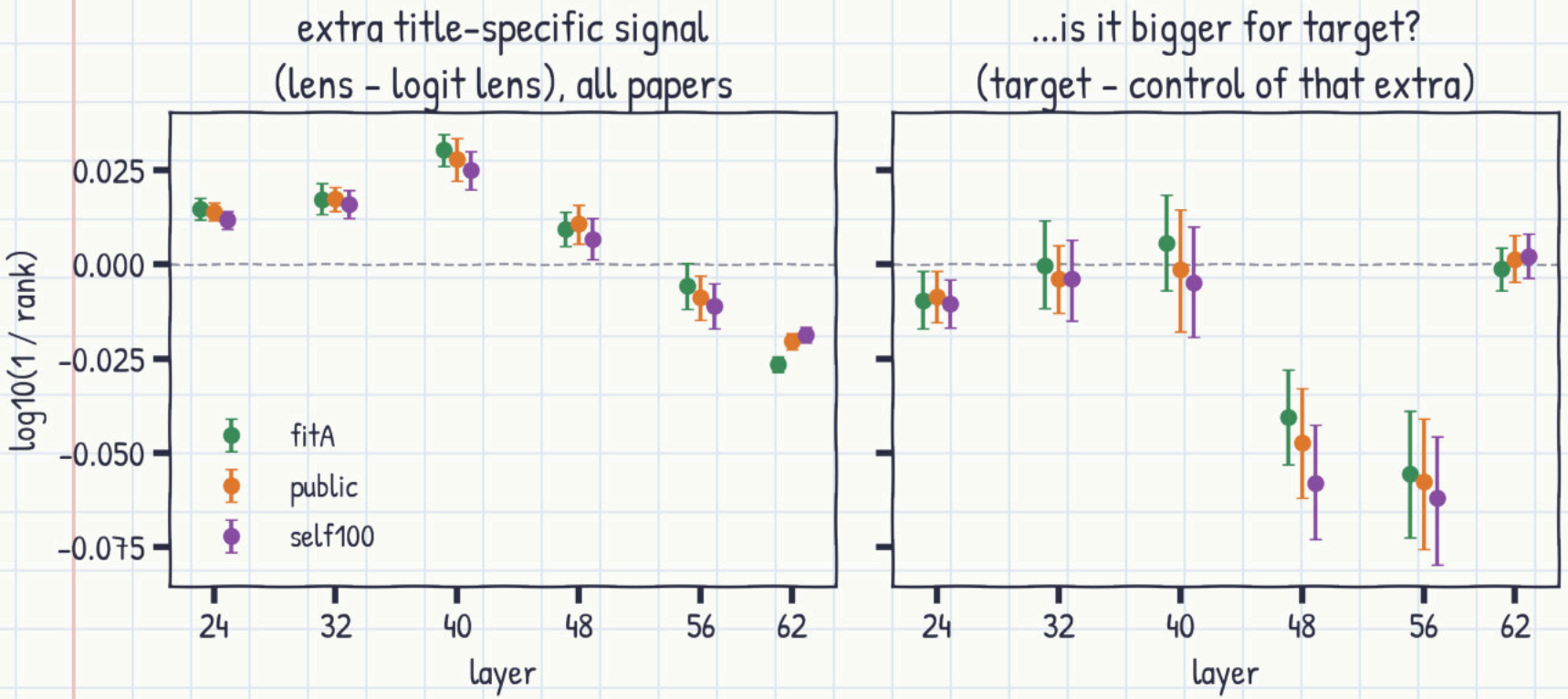
**L62:** logit lens best (59%)

(same, L62)

Stage 10

# Do lenses reveal hidden paper knowledge?

paired differences, 95% CI







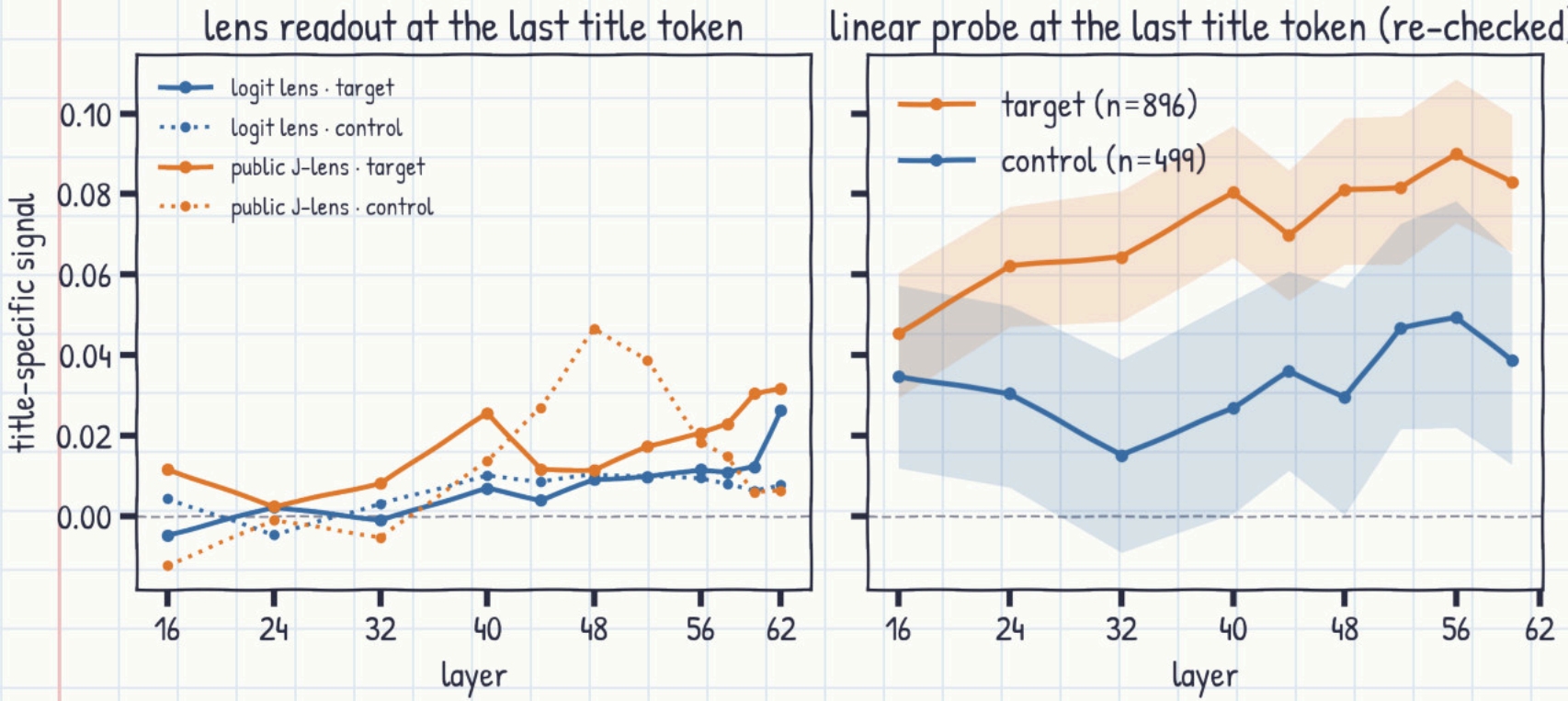
Extra signal: **+0.01 to +0.03** (Stage 10 X3, L24–L48, 9,108 papers, §6.9)

vs ~0.44 at L62 (same, L62)

**Not bigger for target**
(even negative at L48–56)

# Read early: a probe finds it at the title

title-specific signal at the last title token · lenses: all 9,108 papers · probe: re-checked, title reuse removed: 1,863 papers (896 target, 499 control) (Stage 11 E4; Stage 13 E4b)




lenses: **~0.01–0.03** (Stage 11 E4, 4,621 + 2,220, §6.11)

probe: **0.05 -> 0.08** (Stage 13 E4b, 1,863 papers, first name token only)

target > control by **+0.03–0.08** (E4b, 896 vs 499, within bin)

re-checked: team folds, title reuse removed

scores the first name token only (mostly an initial); subtracts false_0 only

**read early** (title, from L24), **spoken late** (names, L57–61)

Probe scores only the first name token (mostly an initial); all-token, by-role version (addendum 9, done): surname first token L40 +0.071 [0.058, 0.085] (1,395 papers; e4b_alltok.md).

Stages 1–9 · 20–200 papers

# What we saw

- Self-100 fits the final distribution better; author ranking not better
- J-Lens − Vanilla ≈0 (7.4: +0.002; Stage 8: −1.4 pp)(50 / 200 papers)
- Stage 9: only Fit A fitted, stability not tested

Stages 10–11 · 9,108 papers

# What scale showed

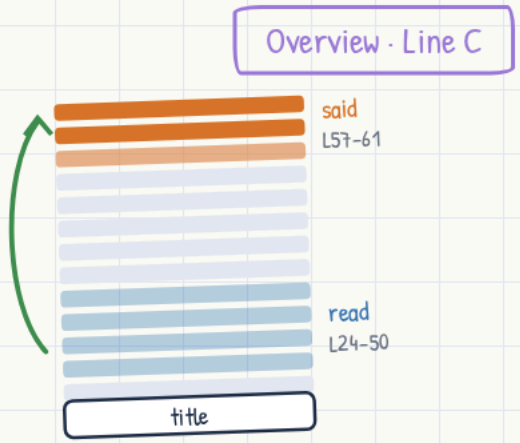
- Jacobian lenses: top-10 rate 15–18% at L48 vs 1% for logit lens(9,108 papers; X3_means.csv)
- Extra title-specific signal only +0.01–0.03(L24–L48; X3_means.csv), not target-specific
- At L62 the plain logit lens is best
- Linear probe at the title token: +0.05 → +0.08 (re-checked) (E4b, scores only the first name token; e4b_trainstd_compare.md), paper-specific

**Line B: the information is there, but not in a form the lenses read**

Both phases agree: no lens reveals hidden paper knowledge in middle layers. Scale adds the positive side: a trained probe can decode it at the title, so the limit is the readout, not the absence of information.

Line C

# When does it appear, and when is it used?




small samples asked

Which layer peaks? (L56 kept standing out)

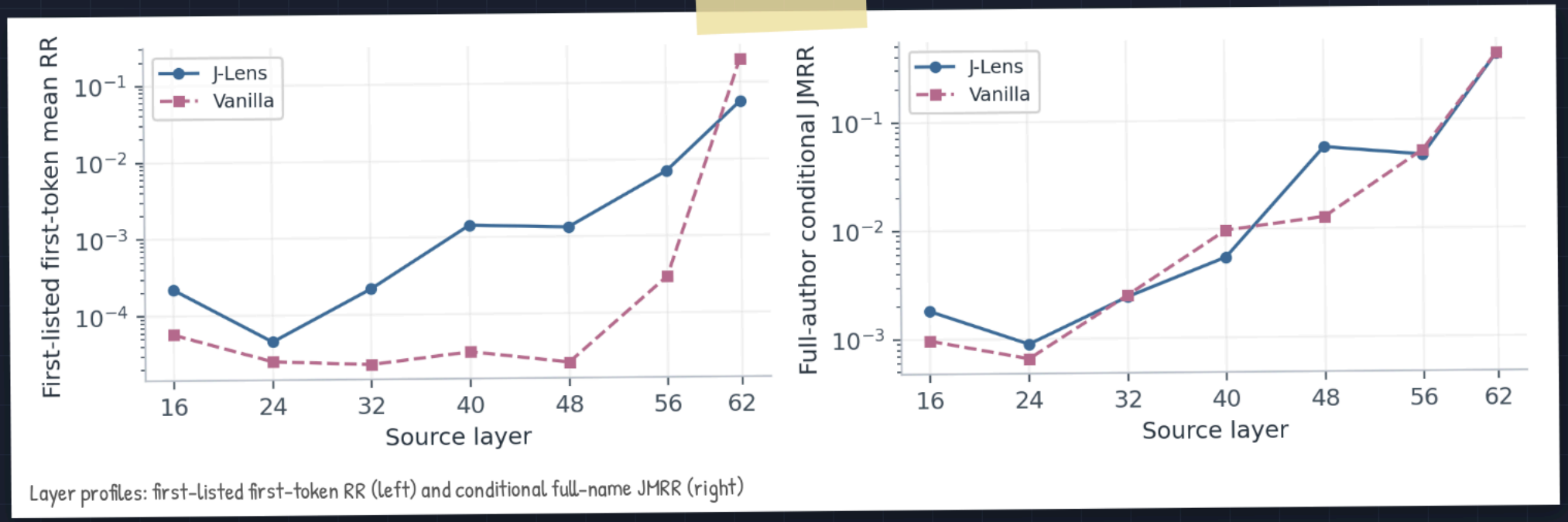
→

the full cohort answered

Readable late (L57–61) but read from the title early (L24–50)

# First readout: is any author signal there?

20 papers, 93 authors (A8 / B8 / C3 / D1); ~91 s per paper, ~29.5 GiB GPU(351_stage4_readout.log)




Layer profiles: first-listed first-token RR (left) and conditional full-name JMRR (right)

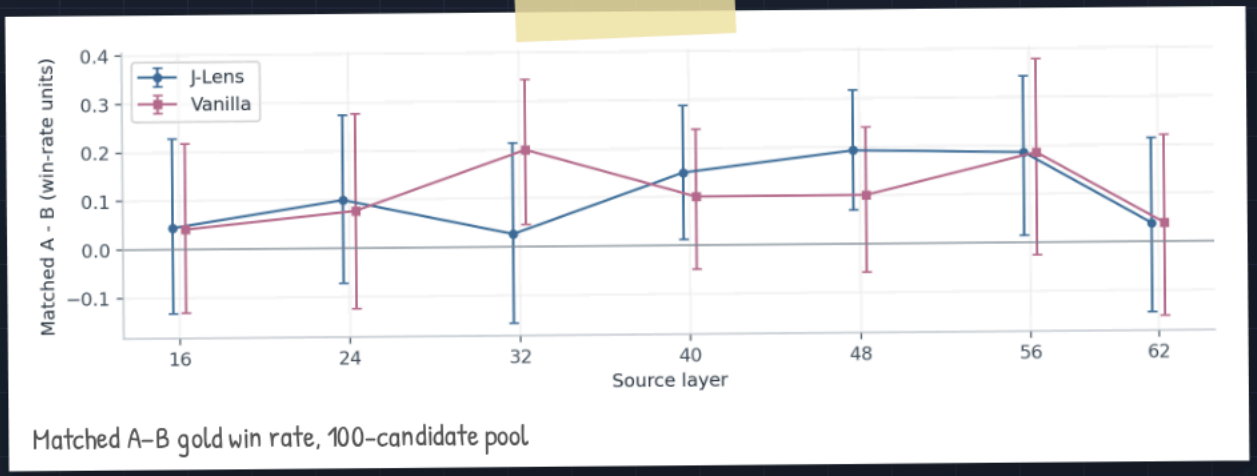
- The pipeline works and J-Lens rises above Vanilla in middle layers
- L56 first-author mean RR 0.0070, median only 0.0009(20 papers; audit_report.md table 1)

**One paper = 71% of the L56 mean**

Dropping it: 0.0070 → 0.0021.(same file) A layer peak can be a single easy paper, so every later stage adds controls.

# Gold vs 100 real false authors

50 matched (same field, token count, year) + 50 random names per paper; 75,000 scored pairs(candidate_pairwise_scores.csv)



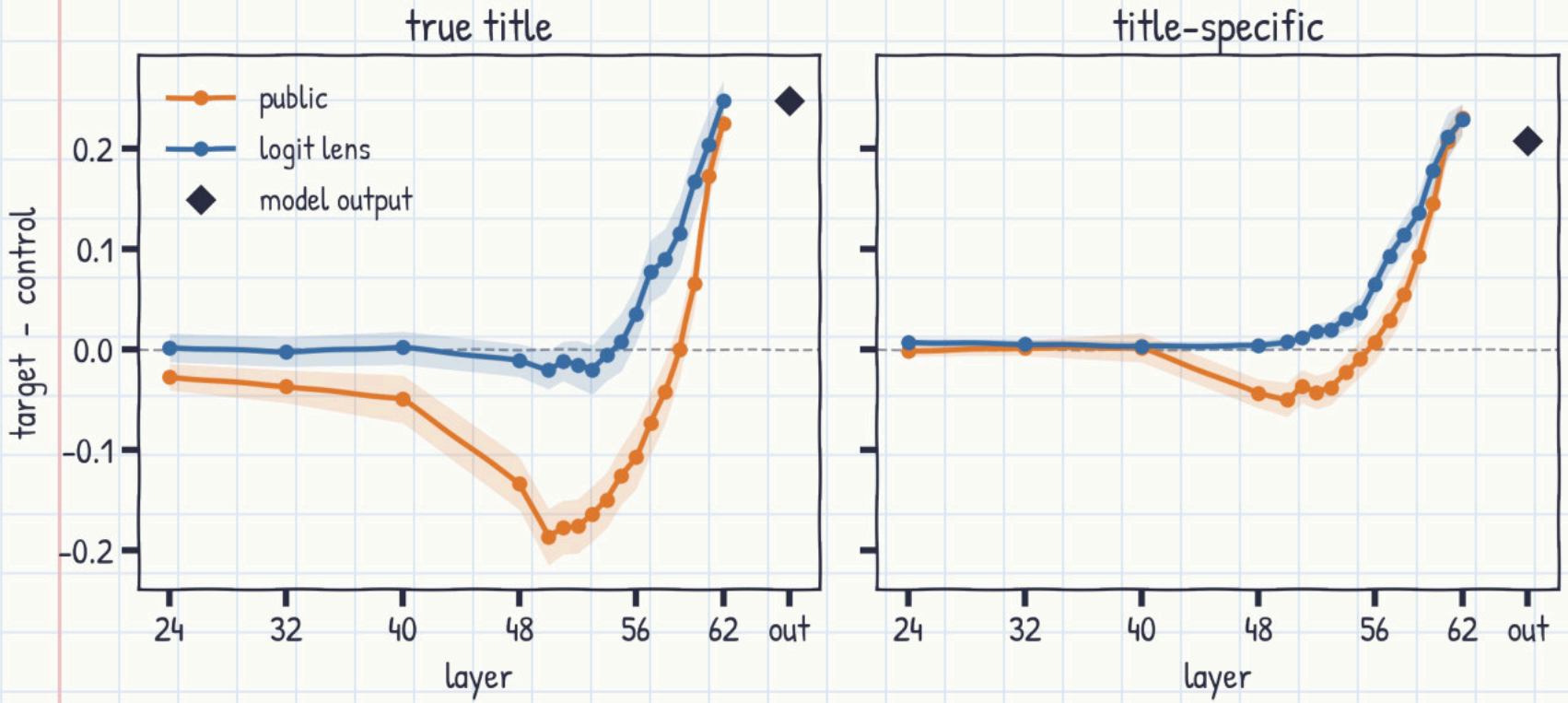

Matched A–B gold win rate, 100-candidate pool

- J-Lens A–B: L40 +0.15, L48 +0.19, L56 +0.19 (CIs above 0); L62 +0.04 (spans 0)(20 pairs, matched_ab_bootstrap.csv)
- 7-layer integrated: +0.105 [−0.012, +0.229](20 pairs; recomputed 10-30)
- 156/5,000 false names share the gold first token (ties) (quality.json)
- 100 names → only ~46 distinct first tokens(mean of 50 papers; candidates csv)

Gold beats real names in middle layers for A more than B. But the same names might win under any title: 7.3 tests that.

# Target pulls ahead only in the last layers

target − control, within citation bin, 95% CI




Middle layers: **no head start**

L62 / output:

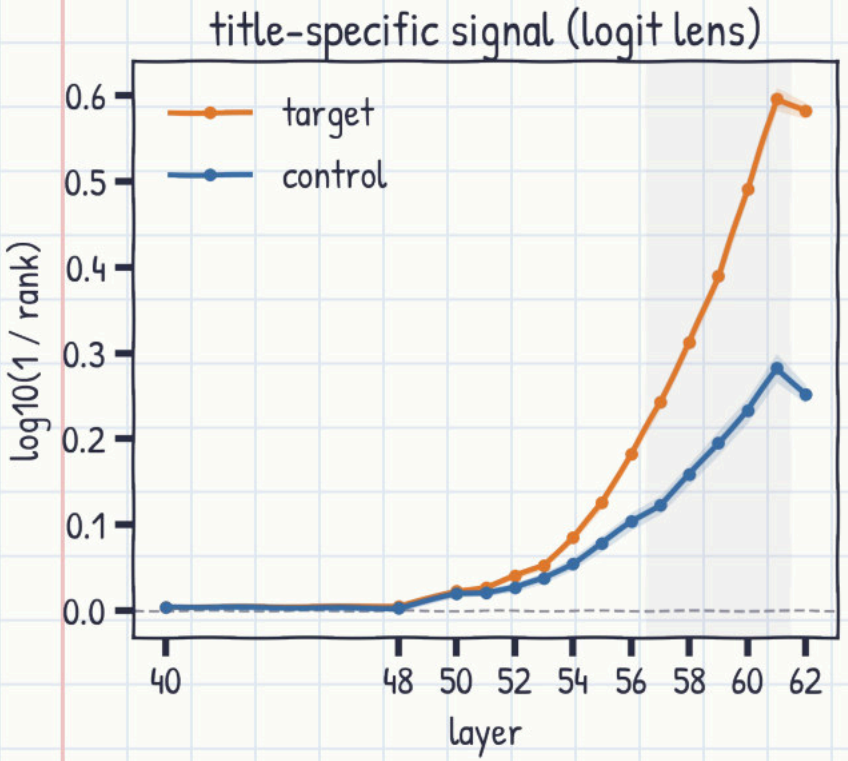
**+0.21 to +0.23** (Stage 10, L62 + final L63, 3,734 + 2,182 papers, §6.4)

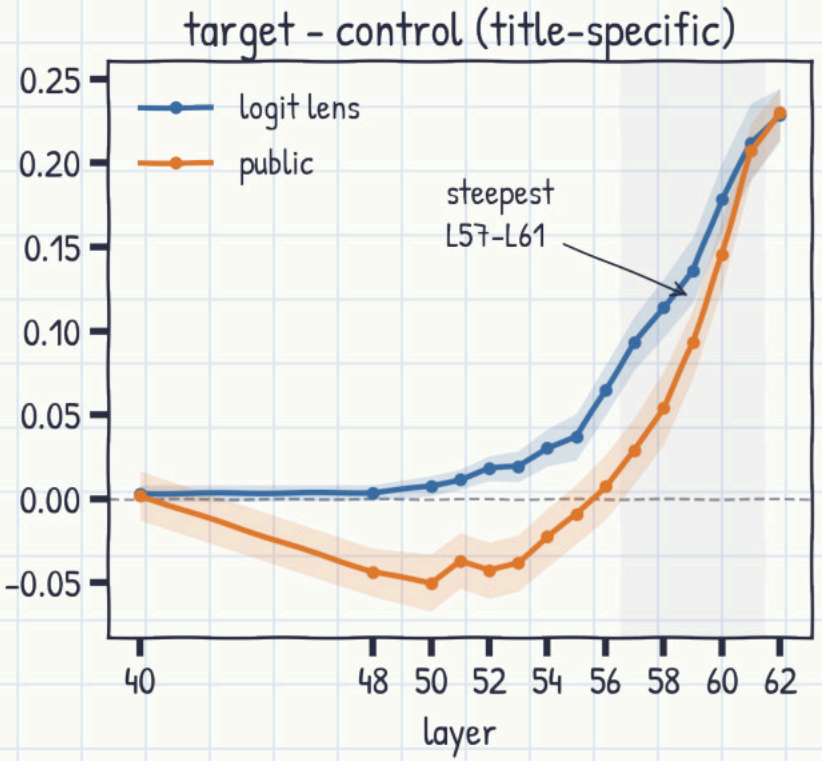
almost all title-specific

**R0 only** (no author written): output +0.199 vs +0.207 (Stage 10, 3,423 + 2,151 R0 papers, §6.13)

# It builds up over the last dozen layers

fine layers L40–L62 · title-specific signal · 95% CI




Logit lens: target - control

**+0.007 at L50**

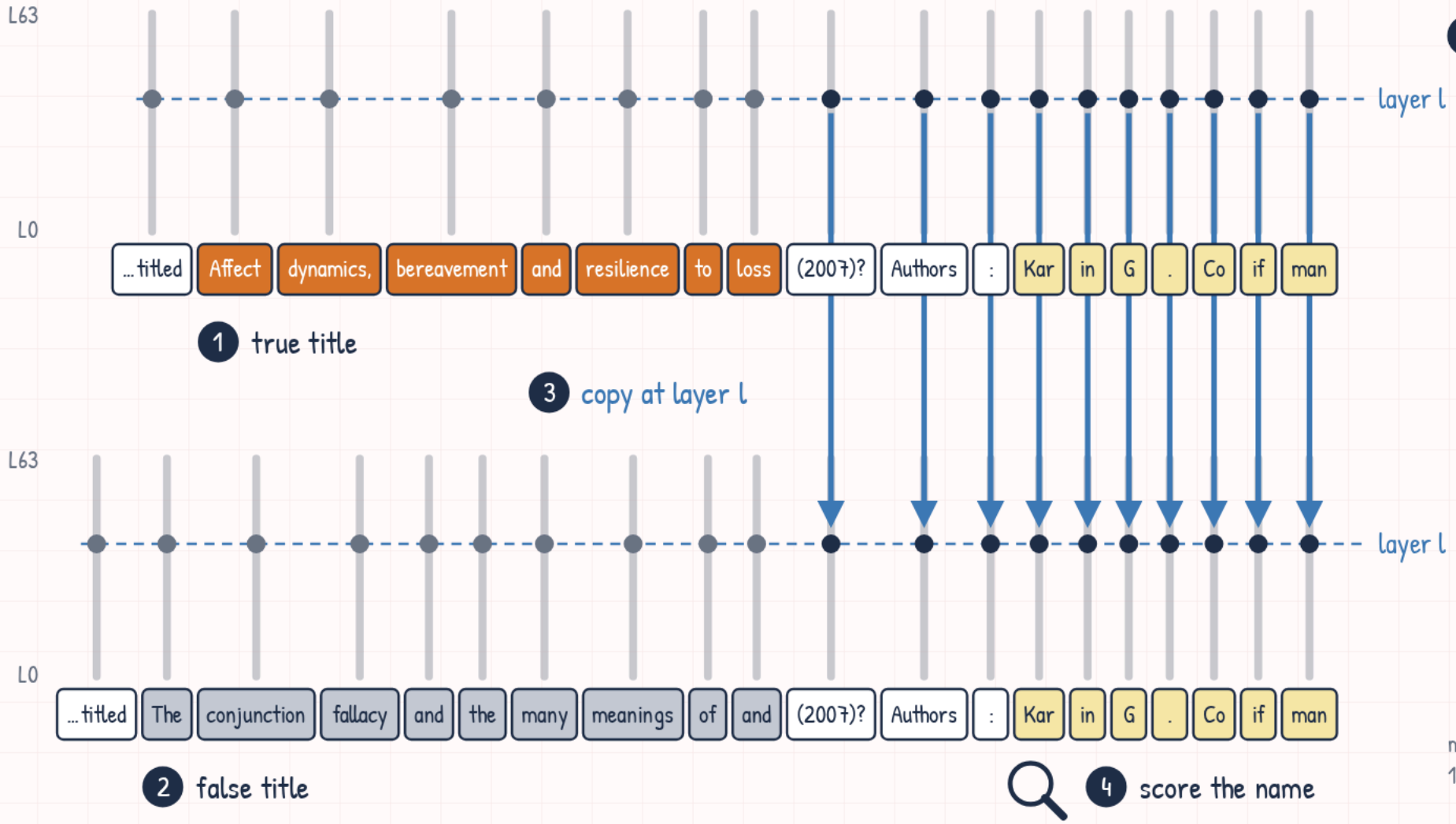
**+0.23 at L62** (Stage 10 logit lens, 3,734 + 2,182 papers, §6.10)

steepest rise **L57–L61**

# Replacement (patching)

Coifman, Bonanno, Rafaeli (2007)

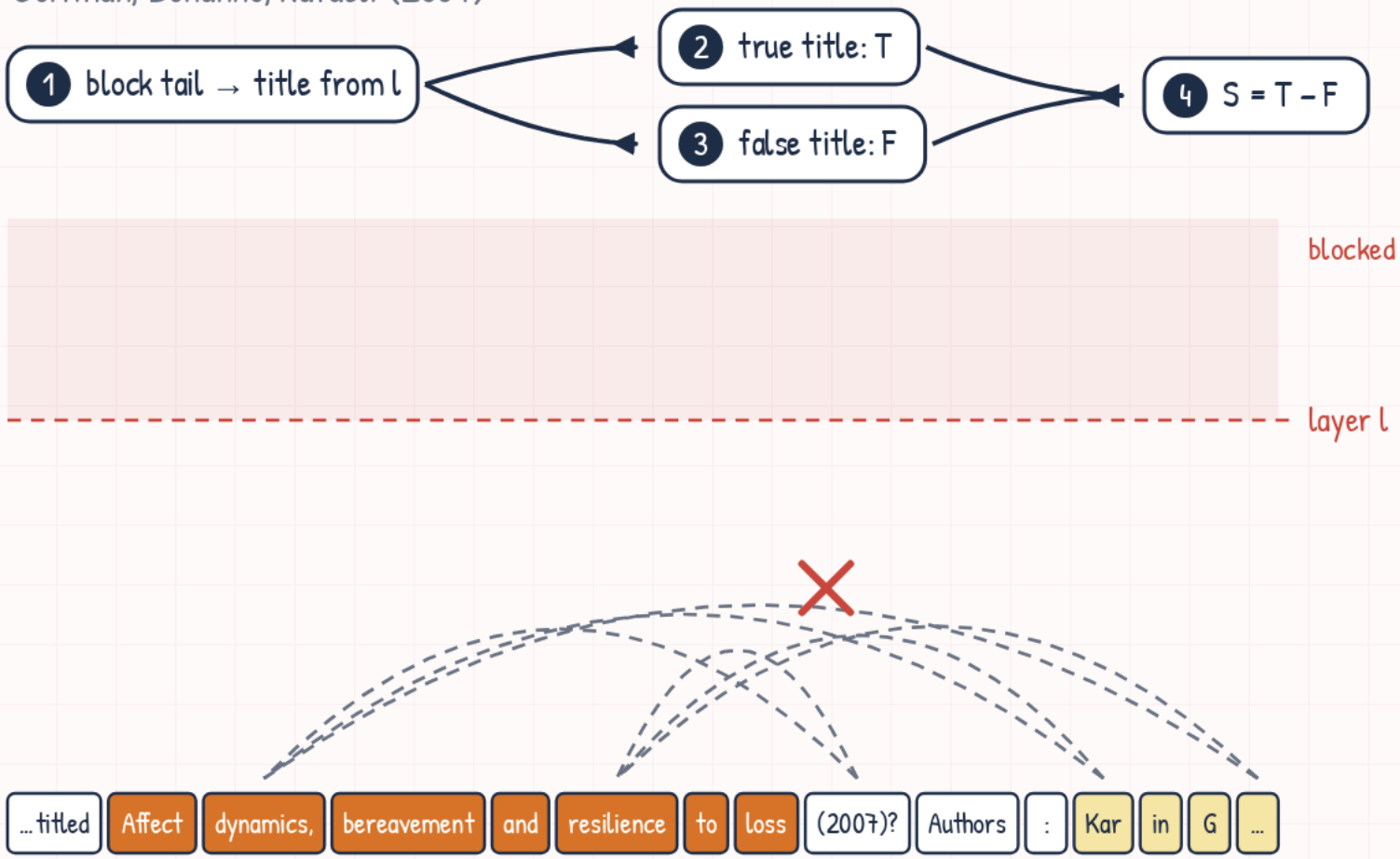
$$R(l) = (\text{patched} - \text{false}) \div (\text{true} - \text{false})$$

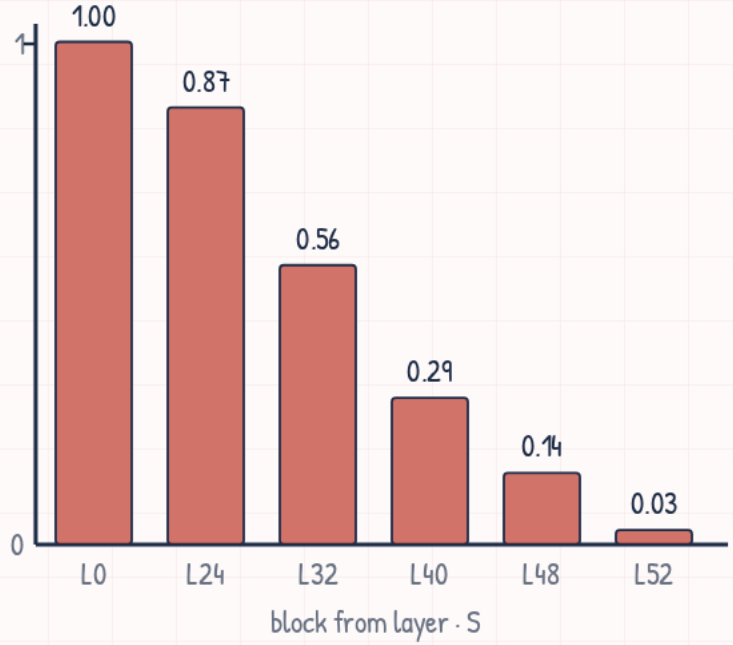



R(l): Stage 13 X6, mean of 5 false titles, all 1,186 papers; 381_stage13/summary_x6/x6_recovery.csv

# Knockout (block attention)

Coifman, Bonanno, Rafaeli (2007)




L52: target 0.018 · control 0.051

**S(l) = T(l) − F(l)**

loss as share of (true − false), 1,186 papers

S(l): Stage 13 X7, 1,186 papers (593 target + 593 control); 381_stage13/reports/x7_report.md

# Stage 11: intervene, do not just read

593 + 593 papers: min(60, n_t, n_c) per citation bin, seed 11 (Stage 11 sample.parquet, §6.11)

patch uses false_0 · knockout masks all heads · all 9,108 for probe and chain (Stage 10 manifest.json)

**patch**

copy the true-title hidden state into the false-title run at one layer

**knock out**

block attention to the title tokens (all heads) from layer l to the last layer

**probe**

linear map from the last title token to the output state

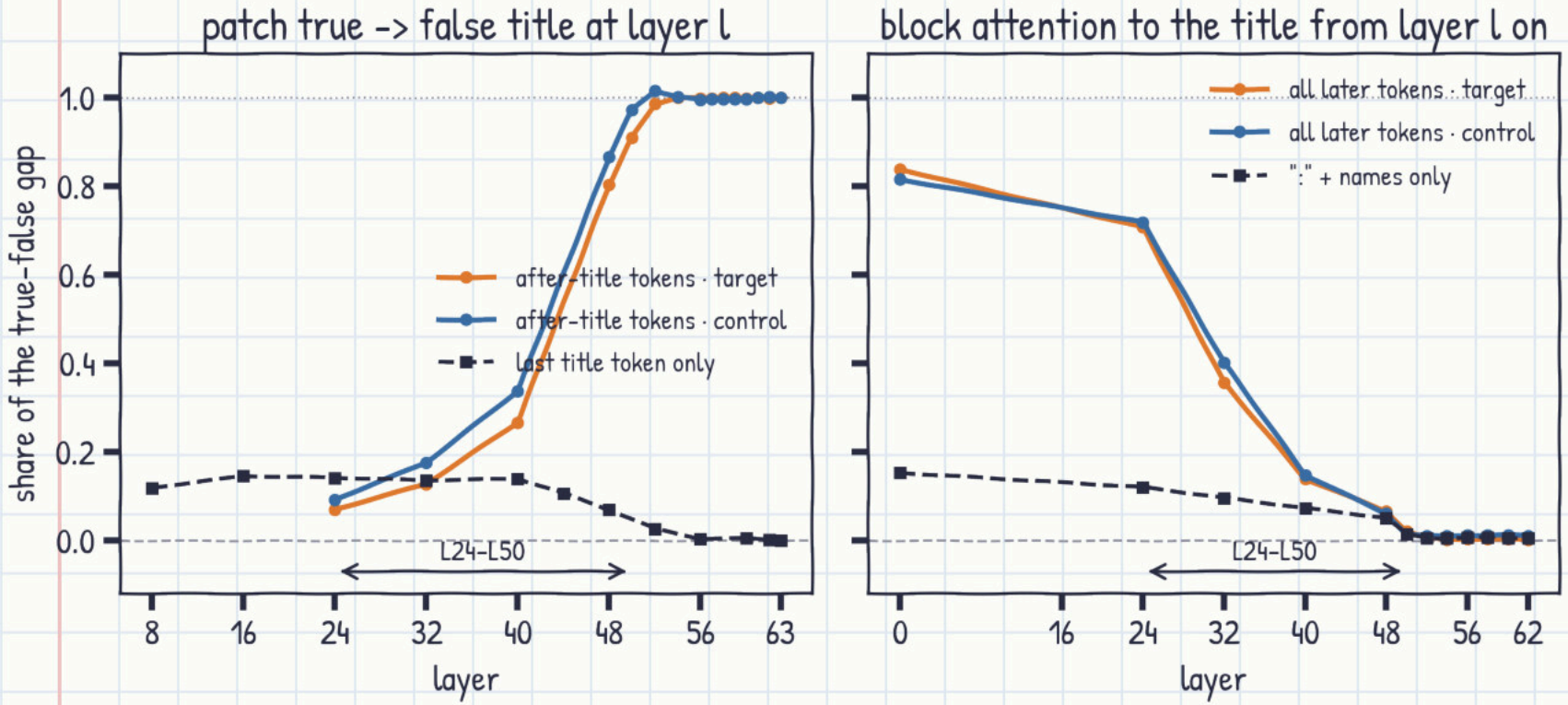
**chain**

"Authors: A1, A2, ..." instead of each author alone

Score: rank of the correct name token at the model output; "share of the true-false gap" = (intervened - false) / (true - false).

# The title is read in L24-L50, early

model output · share of the true-false gap · CIs about +/-0.02 (Stage 11 E1/E2, 593 + 593 papers, §6.11)




L52+: title no longer needed

the information enters the **name positions at L44–52**, partly via the tokens right after the title

same timing for target and control

Stages 1–9 · 20–200 papers

# What we saw

- Stage 4 peak at L56 driven by one paper (71%)(audit_report.md)
- 7.2: A–B win-rate positive at L40–L56
- 7.3, 8: L56 title advantage; L56 became the primary layer

Stages 10–11 · 9,108 papers

# What scale showed

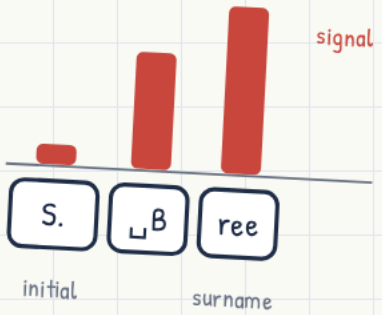
- L56 is where it *starts*; steepest rise L57–L61
- Patching: copying tail states recovers 80–87% at L48, 99.6% from L52(Stage 11 E1, 593 + 593; x6_report.md)
- Knockout: blocking the title from L52 on costs nothing

**Line C: "read early, spoken late"**

L56 was a real landmark, but not the peak. Interventions add what readouts cannot: the title is consumed in L24–50, and the answer only becomes readable at the name a dozen layers later.

Line D

# Which token carries it?




small samples asked

First token or the rest of the name?

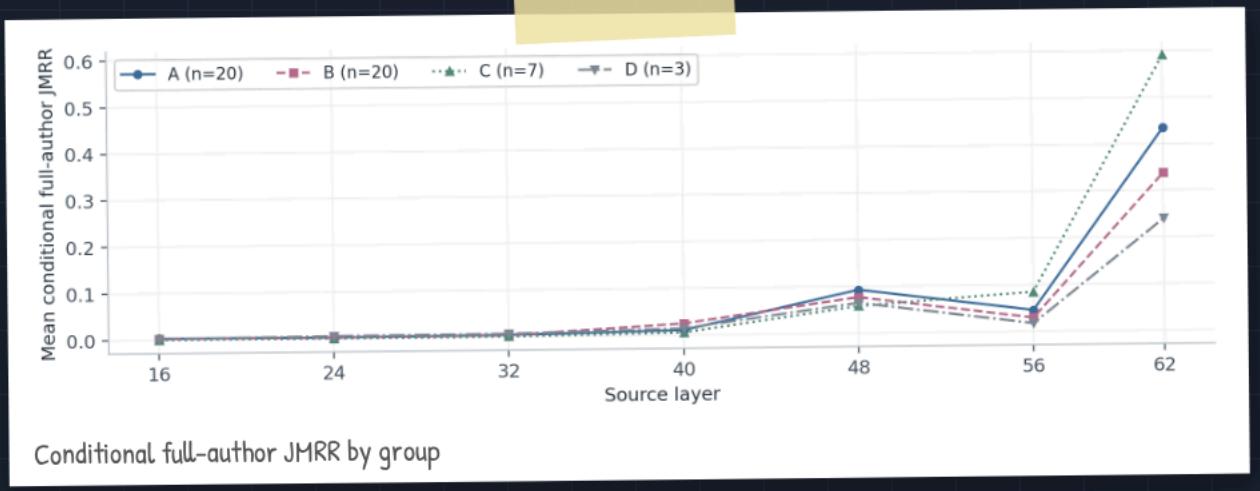
→

the full cohort answered

Depends on name format; the information enters the name positions at L44–52, partly via the tokens right after the title; co-authors partly replace the title

# From first token to the full name

50 papers, 227 authors, 972 name tokens(token_layer_scores.csv); every listed author scored with teacher forcing




Conditional full-author JMRR by group

| Group | Integrated JMRR(integrated_group.csv) |
|---|---|
| A (20) | 0.085 [0.075, 0.096] |
| B (20) | 0.068 [0.057, 0.080] |
| C (7) | 0.106 |
| D (3) | 0.050 |

L62 A–B +0.097 [0.028, 0.163]; Vanilla also +0.091(20 pairs, matched_ab_bootstrap.csv)

Positive, but each token after the first is given the correct prefix. Next: split first token from the rest (7.1).

# The A–B gap sits on the rest of the name

CPU-only re-analysis of Stage 6; 10,000 pair-bootstrap draws




Integrated A–B JMRR by token position

| Endpoint | A–B (7 layers) (20 pairs; layer_and_integrated_effects.csv) |
|---|---|
| First token | +0.001 [−0.011, +0.015] |
| Continuation | +0.021 [+0.002, +0.039] |
| All tokens | +0.017 [+0.003, +0.030] |

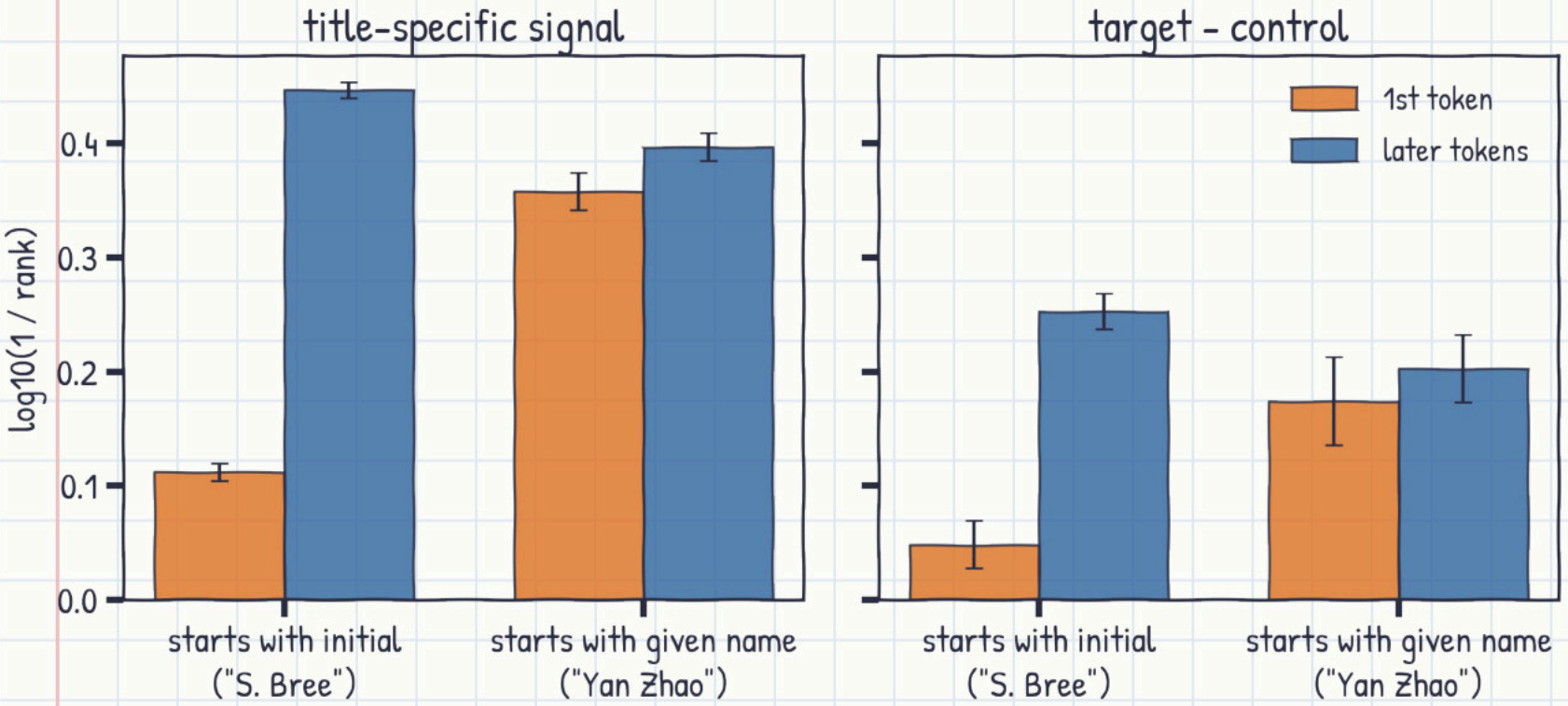
At L62: first +0.014 (12/20 pairs) vs continuation +0.121 (15/20) (layer_position_effects.csv)

With a title contrast (addendum 13 C-1): holds at the output (+0.244) and at L62 (+0.278 logit, +0.317 public J-lens); the public J-lens 6-layer mean is not significant (+0.077 [−0.021, +0.180]) (20 pairs, rest of the name; c13_C1.md)

Completing a name is easier than starting it. A continuation gap cannot show unaided recall. **Open:** is this about initials vs surnames?

# First token or the rest of the name?

model output · split by name format

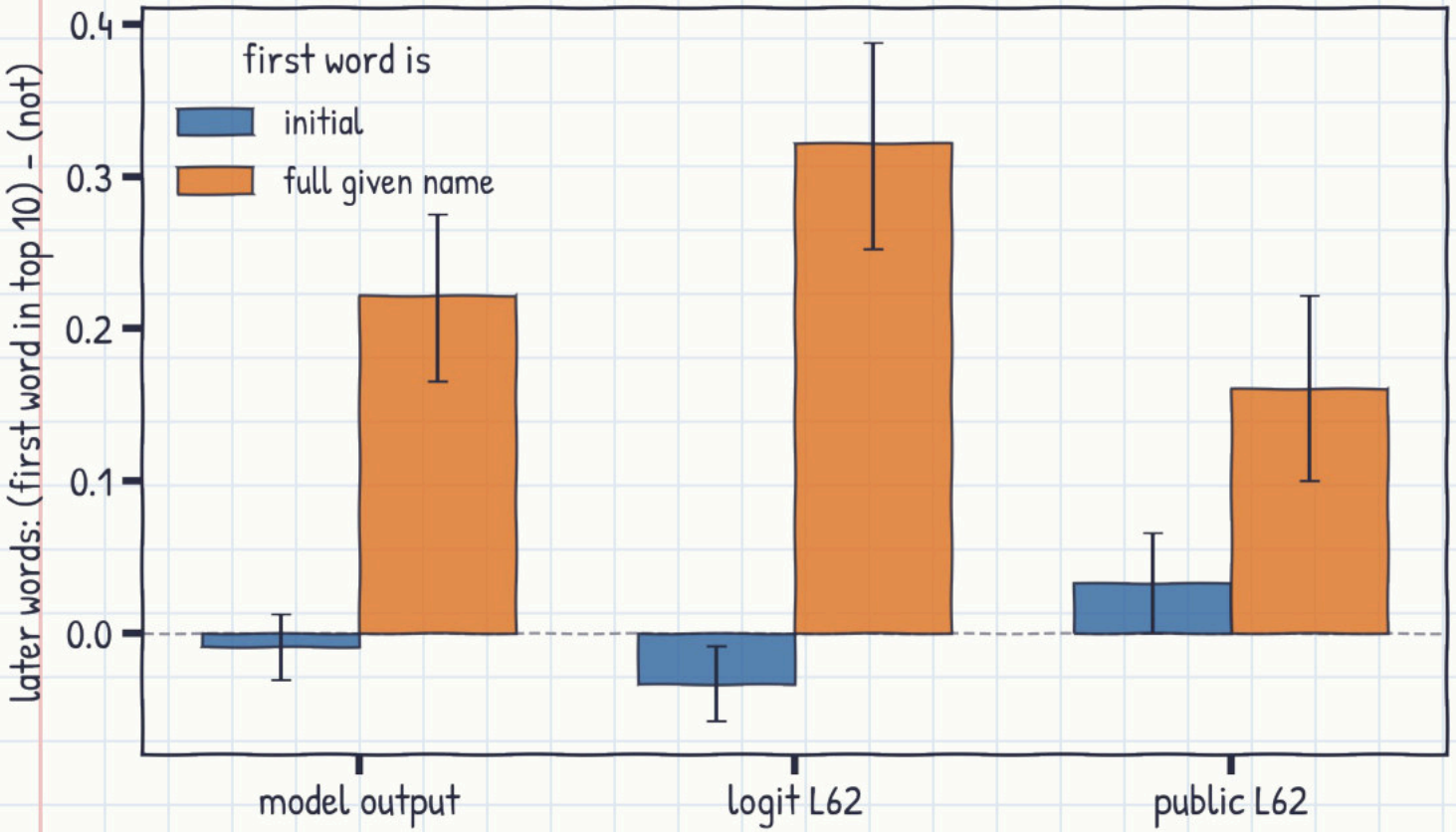



**Initial:** carries little; signal is on the surname

**Full given name:** carries about as much as the surname

# Knowing the given name helps with the surname

true title · later words when the first word is / is not in the top 10




**Given name in top 10**

surname +0.22 (output) (Stage 10, 8,876 authors, §6.7)

+0.32 (logit L62) (same)

**Initial in top 10**: no help